Regular Expressions for Real-World Text Processing

DSA0311 - Natural Language Processing



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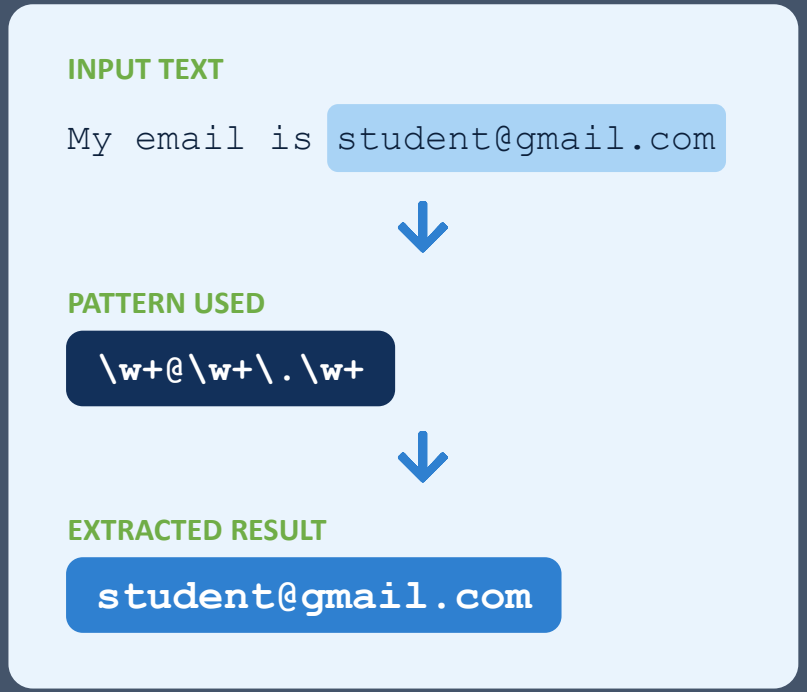
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Introduction to Regular Expressions

Regex = Regular Expression
Regex is the short form of regular expressions.

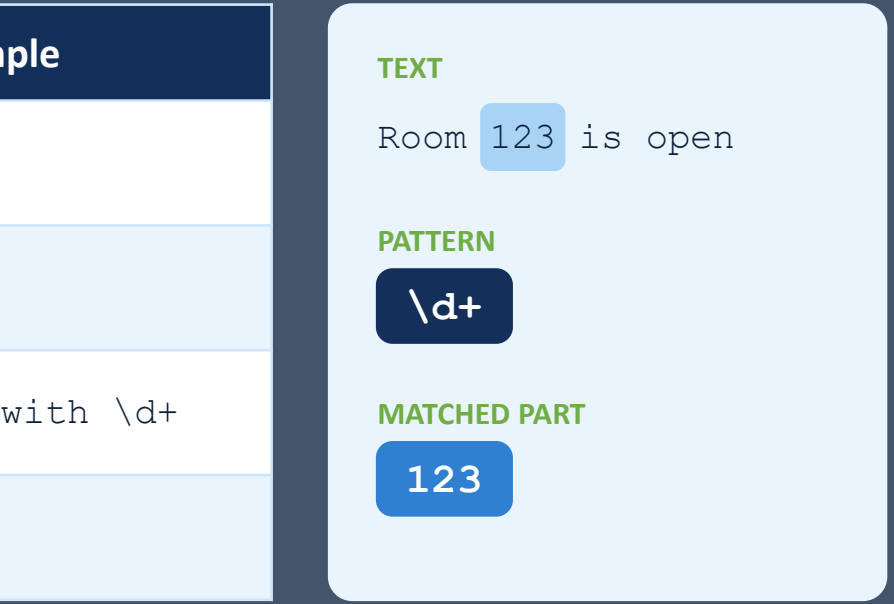
A pattern for text
Used to search, match, validate, and extract text.

Important in NLP
Finds useful data inside raw, messy text.



Basic Regex Symbols

Symbol	Meaning	Example
`\d`	Matches a digit	`5`
`\w`	Matches a word character	`A`
`+`	Matches one or more repetitions	`123 with \d+`
`@`	Matches the literal @ symbol	`abc@`



Combining symbols creates a pattern for finding specific types of text.

Real-World Applications



Email Extraction

Find emails in documents

Example: `student@gmail.com`



Phone Number Detection

Detect phone numbers in text

Example: `9876543210`



Date Extraction

Pick out dates from reports

Example: `22/10/2026`



Keyword Searching

Search keywords in big files

Example: `NLP`

Extracting Numbers Using Python

```
import re

text = "I have 2 books and 15 pens."

result = re.findall(r'\d+', text)

print(result)
```

Output: `['2', '15']`

\d+ matches one or more consecutive digits

1. Import the **re** module
2. Store the input sentence
3. Use **re.findall()** to find all digit sequences
4. Display the extracted numbers

Advantages and Limitations

Advantages

- Fast text searching
- Easy information extraction
- Reduces manual work
- Useful for text validation

Limitations

- Complex patterns can be difficult to write
- Unusual text formats may not match the pattern
- Regex alone cannot fully understand the meaning or context of language

Conclusion

- Regular expressions identify patterns in text.
- They help extract useful information automatically.
- They are useful in Python and NLP applications.
- They make many text-processing tasks faster and easier.



Thank You!

Geo Tag Technical Presentation

